

APPLIED MATHEMATICS-II (MA232432)

Exercise No-1

Question

Do the following activities.

- i. Draw the graphs of the parabolas $(y - k)^2 = 4a(x - h)$ and $(x - h)^2 = 4a(y - k)$ for the values of $a = 2$, $h = 5$ and $k = 2$. Determine the vertex, focus, axis, directrix and latus-rectum of each parabola and mark them on the graphs.
- ii. Draw the graphs of the ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ for the values of $a = 3$, $b = 2$, $h = 1$ and $k = 2$. Determine the eccentricity, centre, foci, vertices, major axis, minor axis, directrices and latus-rectums and mark them on the graph.

Aim:

- i. To Draw the graphs of the parabolas $(y - k)^2 = 4a(x - h)$ and $(x - h)^2 = 4a(y - k)$ for the values of $a = 2$, $h = 5$ and $k = 2$. To Determine the vertex, focus, axis, directrix and latus-rectum of each parabola and mark them on the graphs.
- ii. To Draw the graphs of the ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ for the values of $a = 3$, $b = 2$, $h = 1$ and $k = 2$. To Determine the eccentricity, centre, foci, vertices, major axis, minor axis, directrices and latus-rectums and mark them on the graph.

Procedure:

i. Parabola:

S.No	Description	Geogebra Keywords/Tools
1	Parabola	eq1 : $(y-k)^2 = 4a(x-k)$
2	Vertex V	Vertex(eq1)
3	Focus F	Focus(eq1)
4	majorAxis	MajorAxis(eq1)
5	Directrix	Directrix(eq1)
6	Latus-rectum	➤ Perpendicular Line(Focus, majoraxis) ➤ Intersect (Perpendicular line, eq1)
7	Length of Latus-rectum	➤ Segment (A, B)
8	Other Parabola	➤ input box ➤ caption to Equation ➤ link object to $(y-k)^2 = 4a(x-h)$ ➤ Enter $(x-h)^2 = 4a(y-k)$ in linked object

ii. Ellipse

S.No	Description	GeoGebra Keywords/Tools
1	Ellipse	eq1 : $(x-h)^2/a^2 + (y-k)^2/b^2 = 1$
2	Eccentricity	Eccentricity(eq1)
3	Center	Center(eq1)
4	Vertices	Vertex(eq1)
5	Foci	Focus(eq1)
6	Major axis	MajorAxis(eq1)
7	Minor axis	MinorAxis(eq1)
8	Directrices	Directrix(eq1)
9	Latus-rectums	➤ Perpendicular line(Focus F, majoraxis) ➤ Intersect (Perpendicular line, eq1) ➤ Perpendicular line(Focus G, majoraxis) ➤ Intersect (Perpendicular line, eq1)
10	Length of Latus-rectums	➤ Segment (H,I) ➤ Segment (J,K)

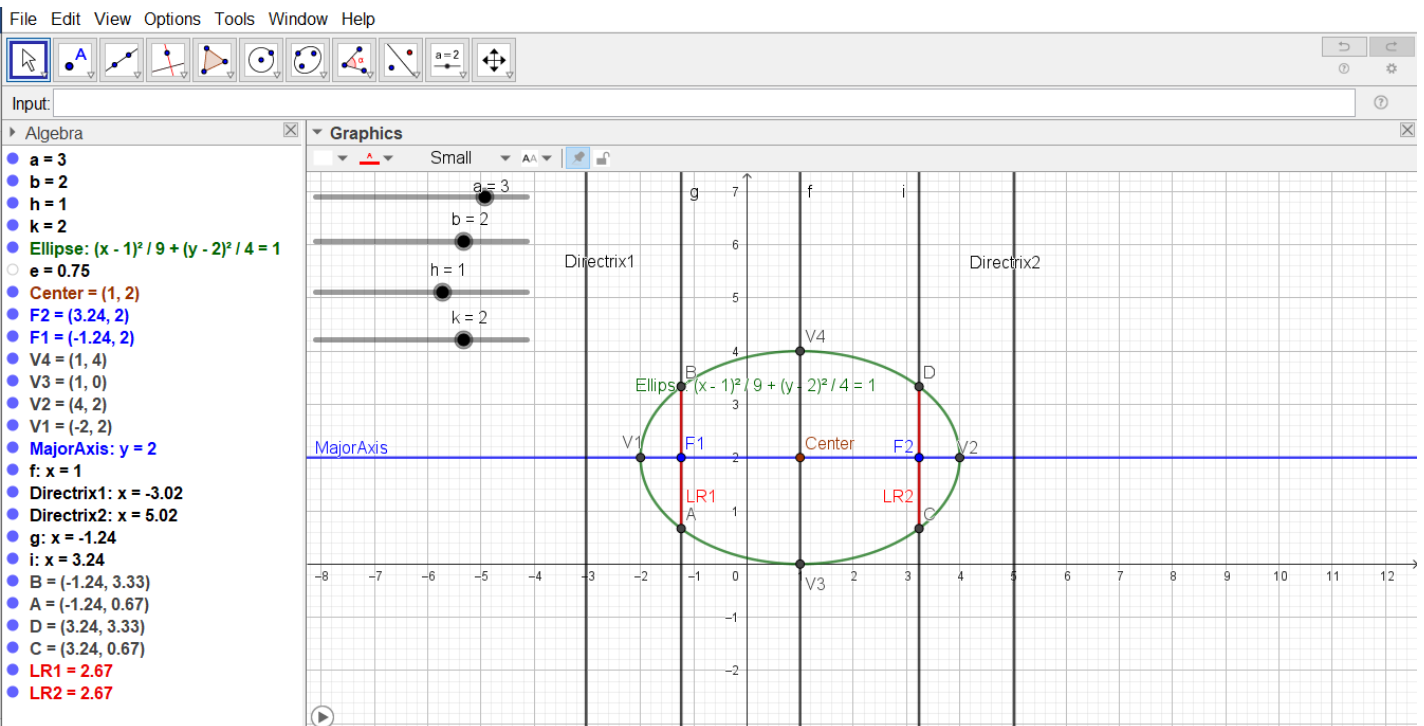
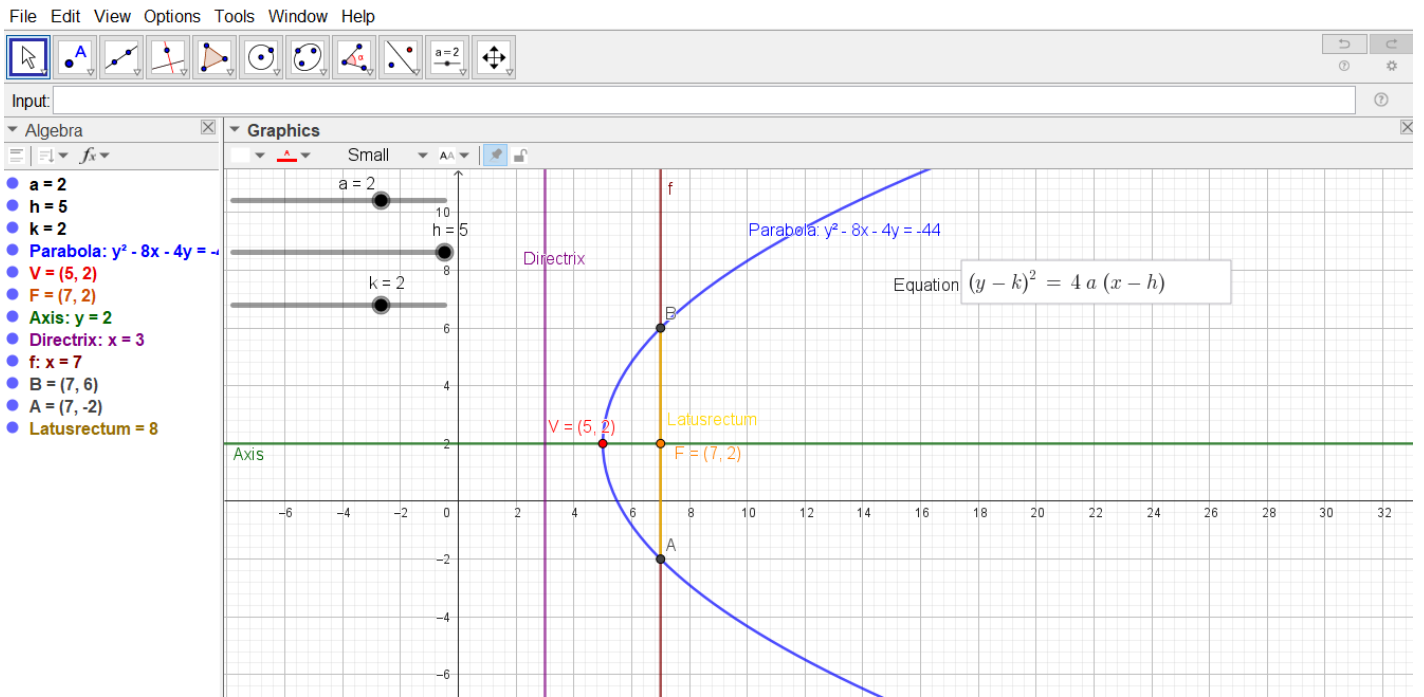
Output:**Table-1** Output for parabolas

Equation of the Parabola	$(y-k)^2=4a(x-h)$	$(x-h)^2=4a(y-k)$
Vertex	(5 , 2)	(5 , 2)
Focus	(7 , 2)	(5 , 4)
Distance from vertex to focus	2	2
Axis	$y = 2$	$x = 5$
Directrix	$x = 3$	$y = 0$
Latus-rectum	$x = 7$	$y = 4$
Length of latus-rectum	8	8

Table-2 Output for ellipses

Equation of the Ellipse	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
Eccentricity	0.75
Centre	(1 , 2)
Foci	(3.24 , 2) and (−1.24 , 2)
Vertices	(1 , 4) , (1 , 0) , (4 , 2) and (−2 , 2)
Major Axis	$y = 2$
MinorAxis	$x = 1$
Directrices	$x = -3.02$ and $x = 5.02$
Latus-rectums	$x = -1.24$ and $x = 3.24$
Length of latus-rectums	2.67

Output :



Exercise No-2

Question

Do the following activities for the given image of a parabolic shaped satellite dish antenna.

- i. Draw a parabola which fits the given image of the dish antenna.
- ii. Write the equation of the parabola.
- iii. Find the vertex, focus, directrix and latus-rectum and mark them on the graph.
- iv. How far from the vertex should the receiver of the antenna be placed?

Aim:

- i. To Draw a parabola which fits the given image of the dish antenna.
- ii. To write the equation of the parabola.
- iii. To find the vertex, focus, directrix and latus-rectum and mark them on the graph.
- iv. To find the distance between vertex and focus.

Procedure:

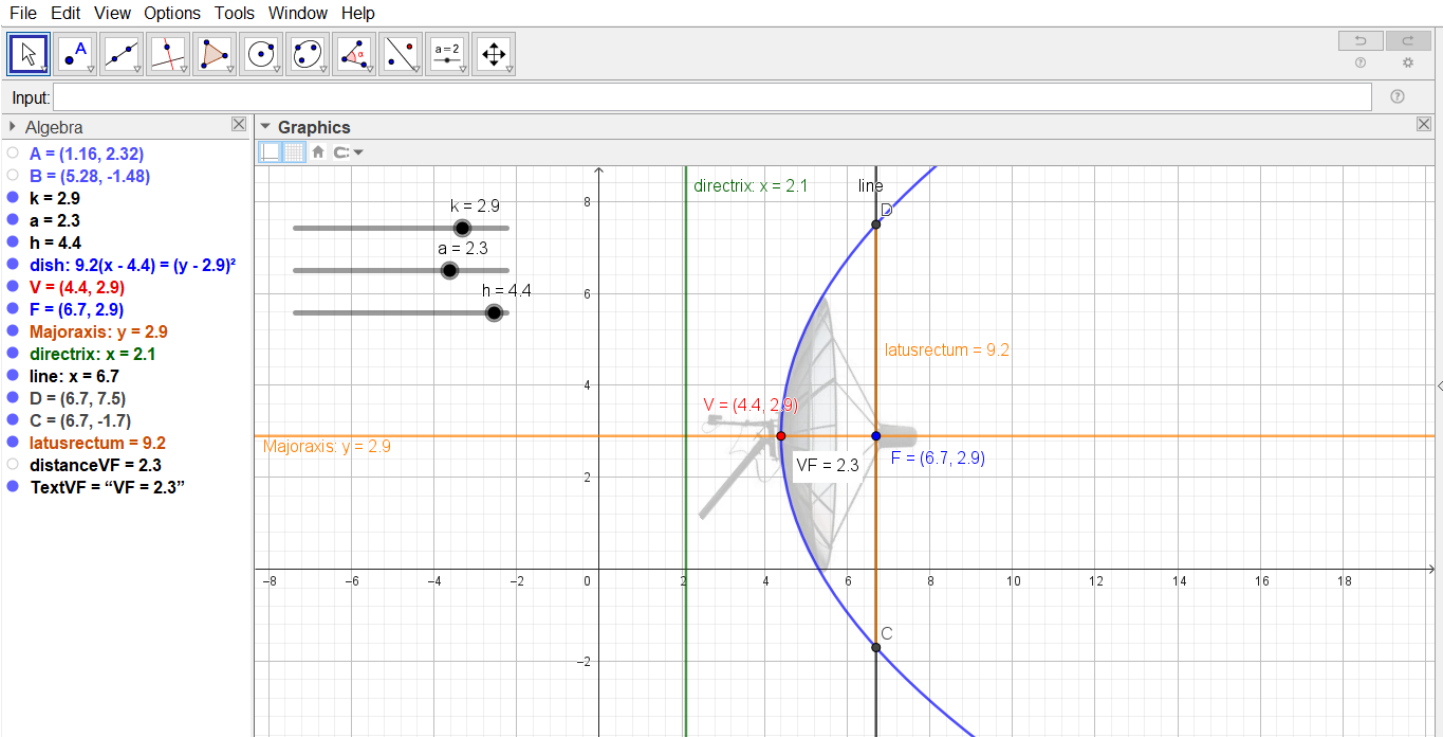
S.No	Description	Geogebra Keywords / Tools
1	Image	Slider tool → image
2	Parabola	eq1 : $(y-k)^2 = 4a (x-k)$
3	Vertex V	Vertex(eq1)
4	Focus F	Focus(eq1)
5	majorAxis	MajorAxis(eq1)
6	Directrix	Directrix(eq1)
7	Latus-rectum	➤ Perpendicular Line(Focus, majoraxis) ➤ Intersect (Perpendicular line, eq1)
8	Length of Latus-rectum (Width)	➤ Segment (C, D)
9	Distance from vertex to focus	➤ Segment (V, F)

Output Table:

Output for parabolic shaped dish antenna

Equation of the Parabola	$(y - 2.9)^2 = 9.2 (x - 4.4)$
Vertex V	(4.4 , 2.9)
Focus F	(6.7 , 2.9)
Directrix	$x = 2.1$
Latus-rectum	$x = 6.7$
Distance from vertex to focus (Distance of the receiver from the vertex)	2.3

Output



Exercise No-3

Question

Do the following activities.

- i. Draw the graphs of $A \sin (Bx + C)$ and $A \cos (Bx + C)$ for some fixed finite real values $A = 5$, $B = 3$ and $C = 2$. Find their domain, range, maximum value, minimum value, amplitude, period and phase shift.
- ii. Draw the graphs of $\sin^{-1} x$ and $\cos^{-1} x$. Find their domain, range, maximum value and minimum value.

Aim

- i. To Draw the graphs of $A \sin (Bx + C)$ and $A \cos (Bx + C)$ for some fixed finite real values $A = 5$, $B = 3$ and $C = 2$. To Find their domain, range, maximum value, minimum value, amplitude, period and phase shift.
- ii. To Draw the graphs of $\sin^{-1} x$ and $\cos^{-1} x$. To Find their domain, range, maximum value and minimum value.

Procedure:

(i)

S.No	Description	Geogebra Keywords/Tools
1	Graph	Graphics $\rightarrow$ X axis $\rightarrow$ distance $\rightarrow \frac{\pi}{2}$ $A \sin (Bx + C)$ $A = 5$, $B = 3$ and $C = 2$
2	Domain	X axis values $\rightarrow (-\infty, \infty)$
3	Range	Y axis values $\rightarrow [-5, 5]$
4	Period	period = $\frac{2\pi}{abs(B)}$
5	Maximum Point	$M = \text{Max}(f, 0, \text{period})$
	Maximum value	maximum = $y(M)$

6	Minimum Point	$m = \text{Min}(f, 0, \text{period})$
	Minimum value	$\text{minimum} = y(m)$
7	Amplitude	$(1/2) (\text{maximum} - \text{minimum})$
8	PhaseShift	$\frac{-C}{\text{abs}(B)}$
9	Input Box	Slider $\rightarrow$ input box For A , B , C and f(x)

(ii)

S.No	Description	Geogebra Keywords/Tools
1	Graph	Graphics $\rightarrow$ Y axis $\rightarrow$ distance $\rightarrow \frac{\pi}{2}$ $\text{asin}(x)$
2	Domain	X axis values $\rightarrow [-1, 1]$
3	Range	Y axis values $\rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$
4	Maximum point	$M = \text{max}(f, -1, 1)$
	Maximum Value	$\text{maximum} = y(M)$
5	Minimum point -- m	$m = \text{min}(f, -1, 1)$
	Minimum Value	$\text{minimum} = y(m)$
6	Input Box	Slider $\rightarrow$ input box for f(x)

Output :

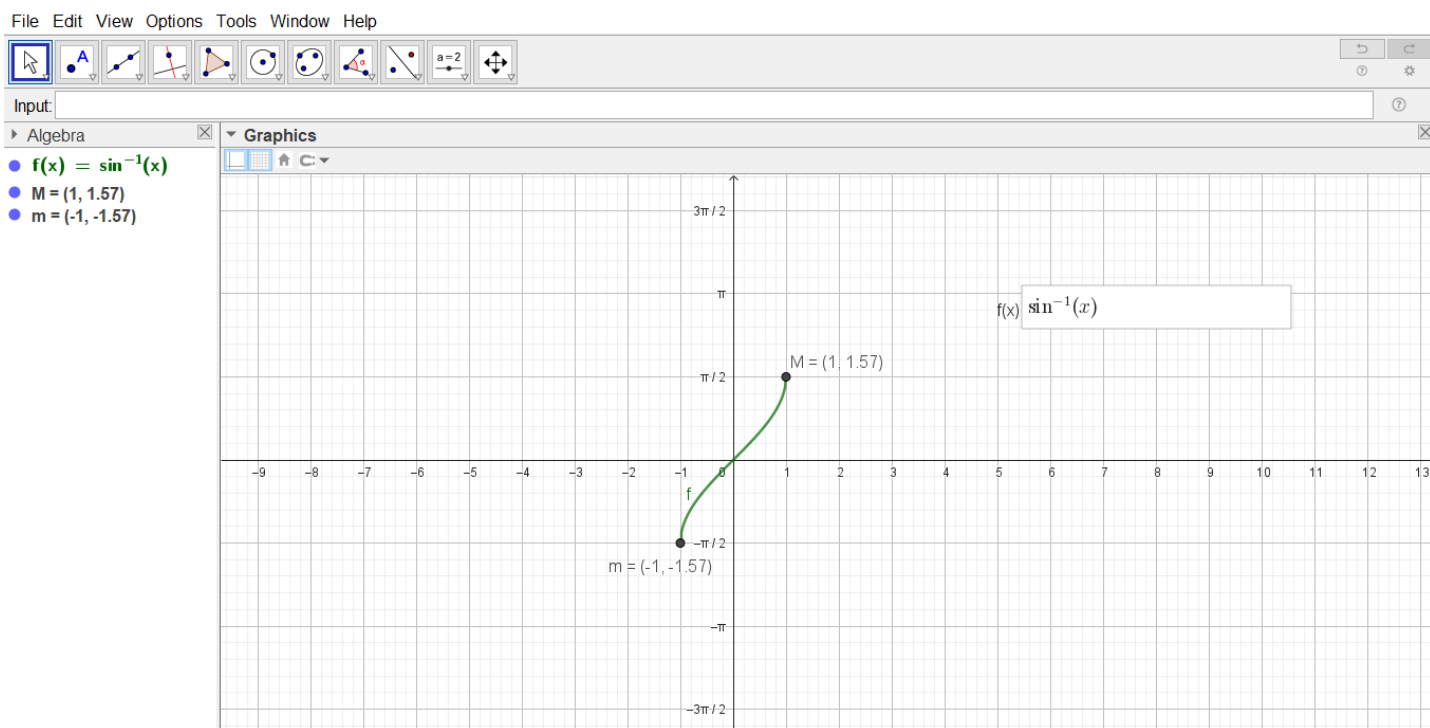
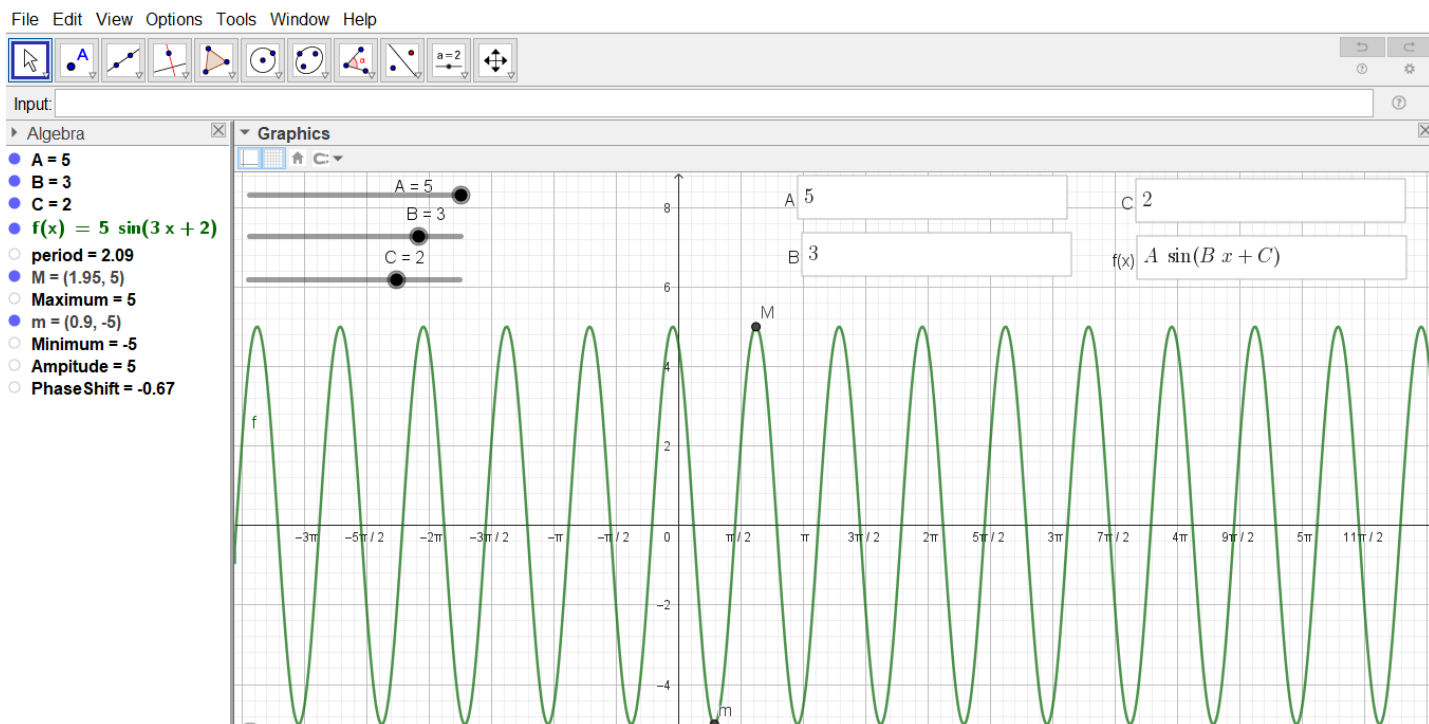
Table-1: Output for waveforms

Description	$y = 5 \sin(3x + 2)$	$y = 4 \cos(2x - 1)$
Domain	$(-\infty, \infty)$	$(-\infty, \infty)$
Range	$[-5, 5]$	$[-4, 4]$
Maximum value	5	4
Minimum value	-5	-4
Amplitude	5	4
Period	2.09	3.14
Phase Shift	-0.67	0.5

Table-2: Output for Inverse Trigonometric Function

Description	$f(x) = \sin^{-1} x$	$f(x) = \cos^{-1} x$
Domain	$[-1, 1]$	$[-1, 1]$
Range	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$[0, \pi]$
Maximum value	1.57	3.14
Minimum value	-1.57	0

Output :



Exercise No-4

Question

The alternating current passing through a circuit is $i(t) = I_m \sin \omega t$, where I_m is the maximum value of the current and ω is the angular velocity. Let R be the resistance and V_m be the maximum voltage. (Take $I_m = 110$, $\omega = 90$ and $R = 5$)

- Graph the sinusoidal waveform of current using the formula $i(t) = I_m \sin \omega t$ for the given values of I_m and ω
- Calculate the maximum voltage V_m using the formula $V_m = I_m R$ for the given value of R
- Graph the sinusoidal waveform of voltage using the formula $v(t) = V_m \sin(\omega t + \frac{\pi}{2})$.
- Determine the value of root mean square (r.m.s) current using the formula $I_{rms} = \frac{I_m}{\sqrt{2}}$
- Determine the frequency using the formula $F = \frac{\omega}{2\pi}$
- Calculate the instantaneous value of the current at t sec.

Aim

- To Graph the sinusoidal waveform of current using the formula $i(t) = I_m \sin \omega t$ for the given values of I_m and ω
- To Calculate the maximum voltage V_m using the formula $V_m = I_m R$ for the given value of R .
- To Graph the sinusoidal waveform of voltage using the formula $v(t) = V_m \sin(\omega t + \frac{\pi}{2})$.
- To Determine the value of root mean square (r.m.s) current using the formula $I_{rms} = \frac{I_m}{\sqrt{2}}$
- To Determine the frequency using the formula $F = \frac{\omega}{2\pi}$
- To Calculate the instantaneous value of the current at t sec.

Procedure:

S.No	Description	Geogebra Keywords/Tools
1	Sinusoidal waveform of $i(t)$	Graphics setting → Basics → Xmin = 0 & Xmax = 0.5 Ymin = -600 & Ymax = 600
		$i(t) = I_m \sin(\omega t)$ Slider → $I_m = 110$, $\omega = 90$ R (Slider) → 5
2	Maximum value of voltage (V_m)	$V_m = I_m R$

3	Sinusoidal waveform of $v(t)$	$v(t) = V_m \sin(\omega t + \frac{\pi}{2})$
4	Root mean square current (I_{rms})	$I_{rms} = I_m / \sqrt{2}$
5	Frequency (F)	$F = \omega / (2\pi)$
6	time	t (slider) Properties → min 0 , max 0.5 , increment =0.1
7	Instantaneous value of current	$I = (t, i(t))$
8	current	current = y(I)

Output

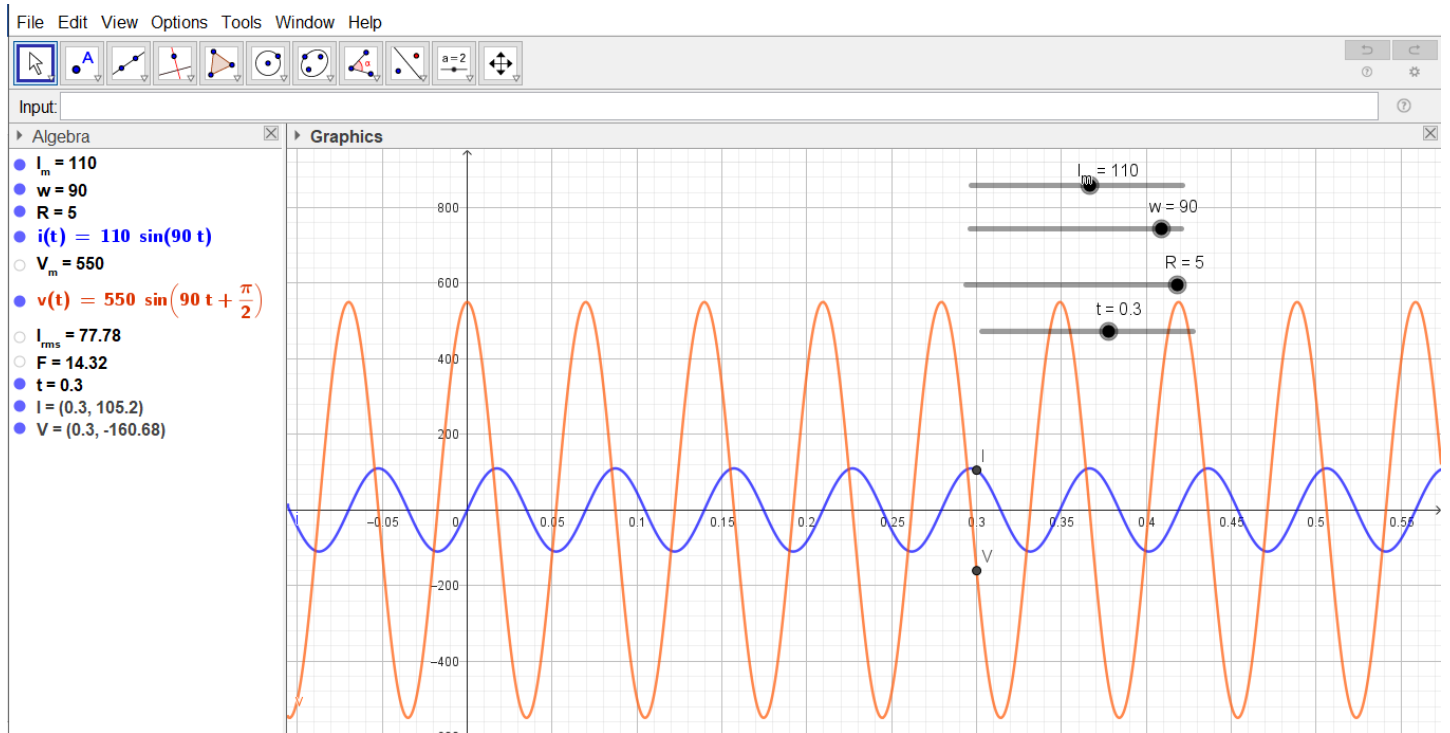
Table-1: Output for current and voltage

Description	Value
Maximum value of current (I_m)	110
Angular velocity (ω)	90
Resistance (R)	5
Maximum value of voltage (V_m)	550
Root mean square current (I_{rms})	77.78
Frequency (F)	14.32

Table-2: Output for instantaneous current for various time values

t	$i(t)$
0	0
0.1	45.33
0.2	-82.61
0.3	105.2
0.4	-109.1
0.5	93.6

Output



Exercise No-5

Question

Do the following activities:

- i. Mark the given complex number $Z=4+3i$ on the Argand plane. Find the real and imaginary parts of Z . Find the distance of Z from x -axis and relate it to the real part of Z . Find the distance of Z from y - axis and relate it to the imaginary part of Z .
- ii. Find the conjugate of Z . Mark Z_1 on the Argand plane. Find the reflection of Z on x -axis and relate it to Z_1 .
- iii. Find the modulus of Z . Find the distance between Z and origin of the Argand plane and relate it to the modulus of Z . Find the modulus of Z_1 and relate it to the modulus of Z .
- iv. Find the argument of Z . Find the angle between the line segment OZ and x axis and relate it to the argument of Z . Find the argument of Z_1 and relate it to the argument of Z .

Aim

- i. To Mark the given complex number $Z=4+3i$ on the Argand plane. To Find the real and imaginary parts of Z . To Find the distance of Z from x -axis and relate it to the real part of Z . To Find the distance of Z from y - axis and relate it to the imaginary part of Z .
- ii. To Find the conjugate of Z . Mark Z_1 on the Argand plane. To Find the reflection of Z on x -axis and relate it to Z_1 .
- iii. To Find the modulus of Z . To Find the distance between Z and origin of the Argand plane and relate it to the modulus of Z . To Find the modulus of Z_1 and relate it to the modulus of Z .
- iv. To Find the argument of Z . To Find the angle between the line segment OZ and x axis and relate it to the argument of Z . To Find the argument of Z_1 and relate it to the argument of Z .

Procedure:

S.No	Description	Geogebra Keywords / Tools
1	Complex number Z	$Z = 4 + 3i$
2	Real part of Z	$RP = \text{real}(Z)$
3	Imaginary part of Z	$IP = \text{imaginary}(Z)$

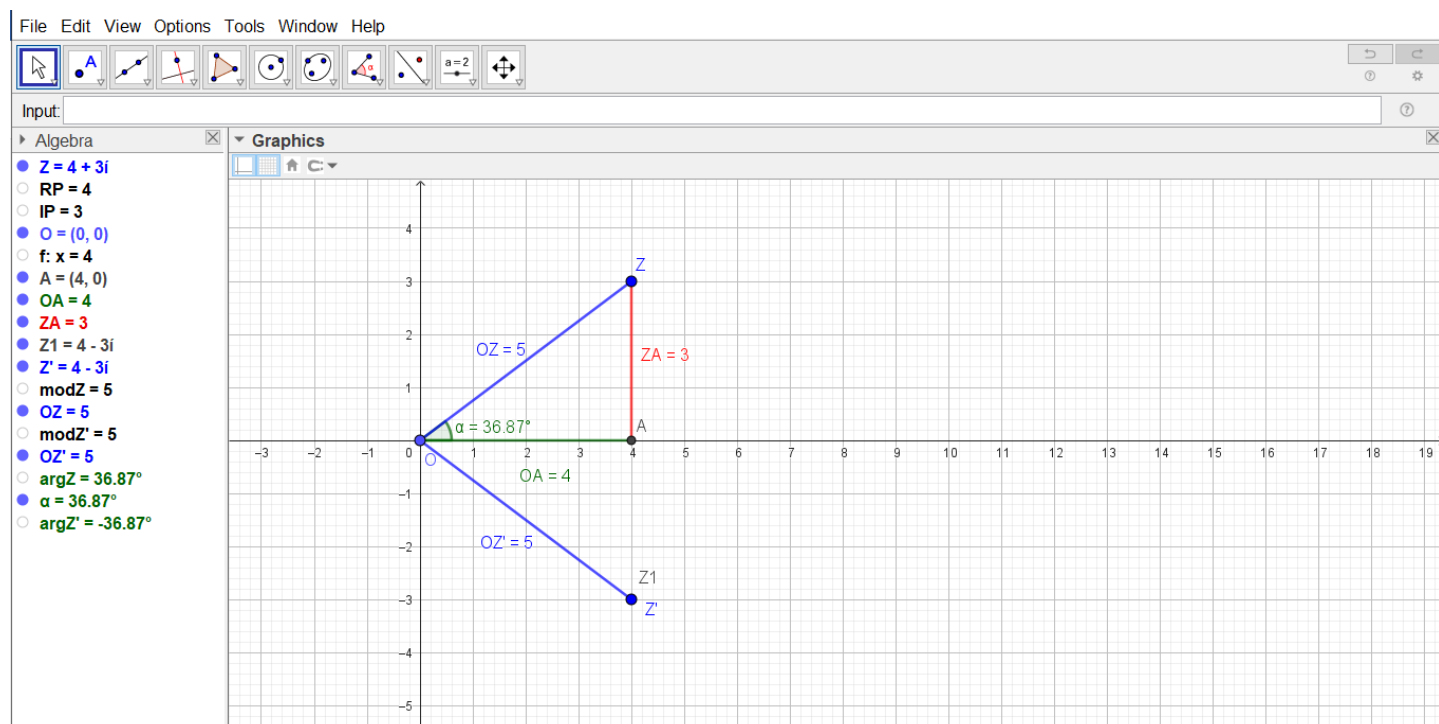
4	Distance of Z from x axis	$O = (0,0)$ Perpendicular line(Z, xAxis) $\rightarrow f$ intersect(f, xAxis) $\rightarrow A$ segment(O,A) $\rightarrow OA$
5	Distance of Z from y axis	segment(Z,A) $\rightarrow ZA$
6	Conjugate of Z	$Z1 = \text{conjugate}(Z)$
7	Reflection of Z	$Z' = \text{reflect}(Z, \text{xAxis})$
8	Modulus of Z	$\text{mod}Z = \text{abs}(Z)$ segment(O,Z) $\rightarrow OZ$
9	Modulus of Z1	$\text{mod}Z1 = \text{abs}(Z1)$ Segment(O,Z1) $\rightarrow OZ1$
10	argument of Z	$\text{arg}Z = \text{arg}(Z)$
11	angle	angle tool $\rightarrow \text{xAxis}, OZ$
12	Argument of Z1	$\text{arg}Z1 = \text{arg}(Z1)$

Output Table : Output for Real Part, Imaginary Part, Conjugate, Modulus and Argument

Complex Number	$Z = 4 + 3i$
Real Part of Z	4
Distance Between x-Axis and Z	4
Are Real Part of Z and Distance Between x-Axis and Z are Equal?	Yes / No
Imaginary part of Z	3
Distance Between y-Axis and Z	3
Are Imaginary Part of Z and Distance Between y-Axis and Z are Equal?	Yes / No
Modulus of Z	5
Distance Between Origin and Z	5

Are Modulus of Z and Distance Between Origin and Z are Equal?	Yes / No
Argument of Z	36.87°
Angle Between x-Axis and the Segment Joining O and Z	36.87°
Are the Argument and Angle Between x-Axis and the Segment are Equal?	Yes / No
Conjugate of Z	$4 - 3i$
Modulus of Z1	5
Argument of Z1	-36.87°
Relationship Between Modulus of Z and Modulus Z1	$\text{mod } Z = \text{mod } Z1$
Relationship Between Argument of Z and Argument of Z1	$\arg Z = -\arg Z1$

Output



Exercise No-6

Question

Do the following activities.

- i. The representation of apparent power as phasor sum of active power and reactive power is given by $S = 550 + 952.63i$. Draw the phasor diagram of the system. Find the numerical value of the apparent power. Also calculate the phase angle and power factor of the system.
- ii. A machine takes 10KW (real power, P) at a power factor of 0.6 from 400V supply. Calculate the total load in KVA(apparent power, S) and KVAR(reactive power, R). Represent the apparent power as a phasor sum of active power and reactive power.

Aim

- i. To Draw the phasor diagram of the system. To Find the numerical value of the apparent power. To calculate the phase angle and power factor of the system.
- ii. To Calculate the total load in KVA(apparent power, S) and KVAR(reactive power, R). To Represent the apparent power as a phasor sum of active power and reactive power.

Procedure:

Window - 1

S.No	Description	Geogebra Keywords / Tools
1	Phasor Sum S	$S = 550 + 952.63i$
2	Phasor Diagram	$O = (0,0)$ $P = (\text{real}(S), 0)$
3	Power	ApparentPower = segment(O,S) ActivePower = segment(O,P) ReactivePower = segment(P,S)
4	Phase angle	$\Phi = \text{angle}(P, O, S)$
5	Power factor	$F = \cos \Phi$

Window - 2

S.No	Description	Geogebra Keywords / Tools
1	Active Power	$P = 10$
2	Power factor	$F = 0.6$
3	Phase angle	$\Phi = \arccos(F)$ (Options → Advanced → Settings → Degree)
4	Voltage	$V = 400$
5	Current	$I = P / (V \cos(\Phi))$
6	Reactive Power	$R = V I \sin(\Phi)$
7	Apparent Power	$S = P + R i$
8	Phasor diagram	$O = (0,0)$ $A = (\text{real}(S), 0)$
9	Power	ApparentPower = segment(O,S) ActivePower = segment(O,A) ReactivePower = segment(S,A)

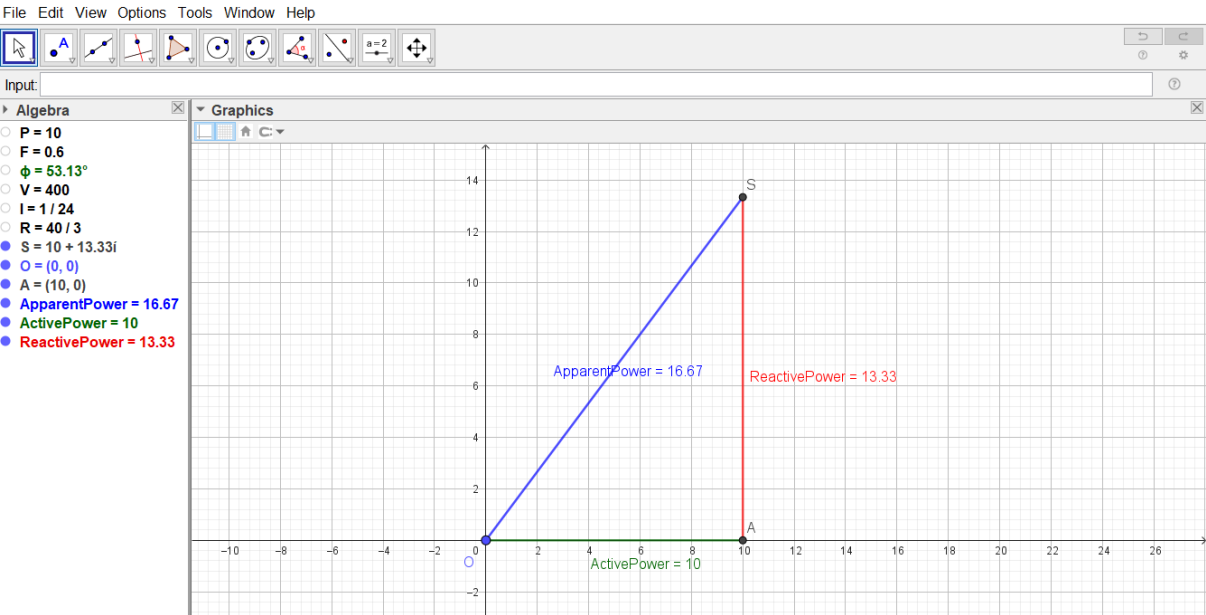
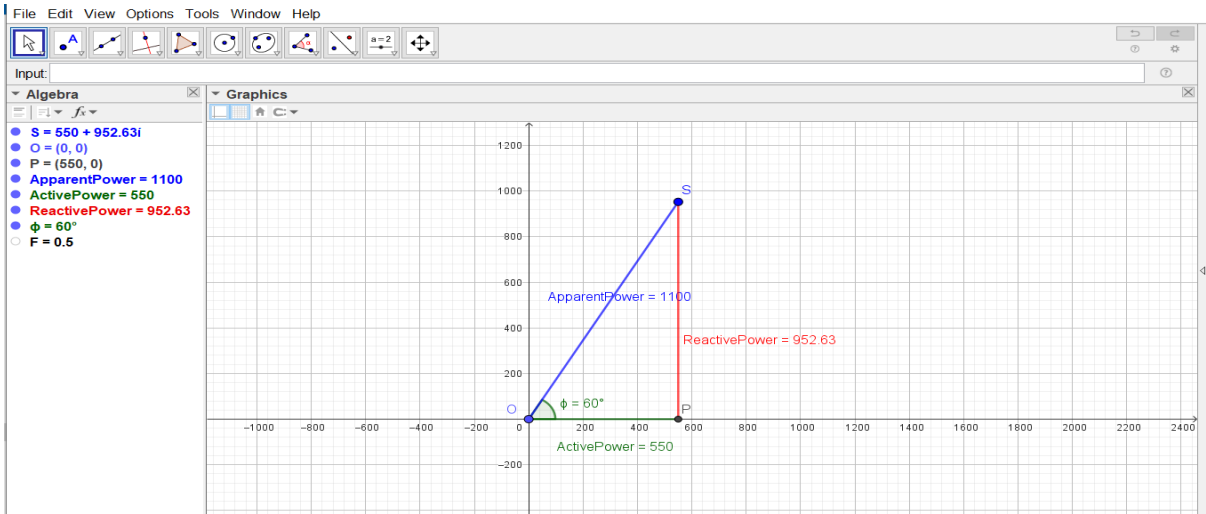
Output Table:**Table-1:** Output for Part-(i)

Description	Value / Expression
Phasor Sum (S)	$550 + 952.63i$
Phase Angle (Φ)	60°
Power Factor (F)	0.5
Apparent Power ($ S $)	1100
Active Power (P)	550
Reactive Power (R)	952.63

Table-2: Output for Part-(ii)

Description	Value / Expression
Active Power (P)	10
Power factor (F)	0.6
Phase Angle (Φ)	53.13°
Voltage (V)	400
Current (I)	$1/24$
Reactive Power (R)	$40/3$
Phasor Sum (S)	$10+13.33i$
Apparent Power ($ S $)	16.67

Output



Exercise No-7

Question

Do the following activities.

- Graph the polynomial function $f(x) = x^2 - 2x + 3$. Find the value of $f(x)$ at $x = 3$ and the limit of $f(x)$ at $x = 3$.
- Graph the rational function $f(x) = \frac{x^2 - 2x + 3}{x + 2}$. Find the value of $f(x)$ at $x = 3$ and the limit of $f(x)$ at $x = 3$.
- Graph the functions $\frac{\sin 3x}{2x}$ and $\frac{\tan 3x}{2x}$. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ and $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$.
- Graph the functions x^3 and $\sin x$. Find their first derivative and second derivative.

Aim:

- To Graph the polynomial function $f(x) = x^2 - 2x + 3$.
Find the value of $f(x)$ at $x = 3$ and the limit of $f(x)$ at $x = 3$.
- To Graph the rational function $f(x) = \frac{x^2 - 2x + 3}{x + 2}$. To Find the value of $f(x)$ at $x = 3$ and the limit of $f(x)$ at $x = 3$.
- To Graph the functions $\frac{\sin 3x}{2x}$ and $\frac{\tan 3x}{2x}$. To Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ and $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$.
- To Graph the functions x^3 and $\sin x$. To Find their first derivative and second derivative.

Procedure:

Window-1

S.No	Description	Geogebra Keywords/Tools
1	$f(x)$	$f(x) = x^2 - 2x + 3$ Options $\rightarrow$ Rounding $\rightarrow$ 4 decimal values
2	$f(x)$ at $x = 3$	$f(a)$ slider $\rightarrow a = 3$ (object Properties $\rightarrow$ min 0, max 10, increment 1)
3	limit of $f(x)$ at $x = a$	LimitPoint = (a , f(a))
		LeftLimitPoint = (a-h , f(a-h)) (slider $\rightarrow h=1$, min 0, max 1)
		RightLimitPoint = (a+h , f(a+h))
		aL = a-h aR = a+h
		LHV = f(a-h) RHV = f(a+h)

4	Input Box	➤ Slider Tool → input box → for $f(x)$, a , h
5	Rational function	➤ Enter $\frac{x^2-2x+3}{x+2}$ in $f(x)$ input box
6	$\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$	➤ Enter $\frac{\sin 3x}{2x}$ in $f(x)$ input box Set $a = 0$ in input box

Window-2

S.No	Description	Geogebra Keywords
1	$f(x)$	$f(x) = \sin x$
2	First derivative	Derivative(f)
3	Second derivative	Derivative(f')
4	Other functions	➤ input box → $f(x)$ Enter other function in input box

Output:

Table-1 Limit of the **polynomial** function $f(x) = x^2 - 2x + 3$

h	a-h	a+h	LHV	RHV
1	2	4	3	11
0.8	2.2	3.8	3.44	9.84
0.6	2.4	3.6	3.96	8.76
0.5	2.5	3.5	4.25	8.25
0.4	2.6	3.4	4.56	7.76
0.2	2.8	3.2	5.24	6.84
0.1	2.9	3.1	5.61	6.41
0.01	2.99	3.01	5.9601	6.0401
0	3	3	6	6
			$\lim_{x \rightarrow a} f(x) = \mathbf{6}$	

Table-2 Limit of the **Rational** function $f(x) = \frac{x^2-2x+3}{x+2}$

h	a-h	a+h	LHV	RHV
1	2	4	0.75	1.833
0.8	2.2	3.8	0.819	1.6966
0.6	2.4	3.6	0.9	1.5643
0.5	2.5	3.5	0.9444	1.5
0.4	2.6	3.4	0.9913	1.437
0.2	2.8	3.2	1.0917	1.3154
0.1	2.9	3.1	1.1449	1.2569
0.01	2.99	3.01	1.1944	1.2056
0	3	3	1.2	1.2
			$\lim_{x \rightarrow a} f(x) = \mathbf{1.2}$	

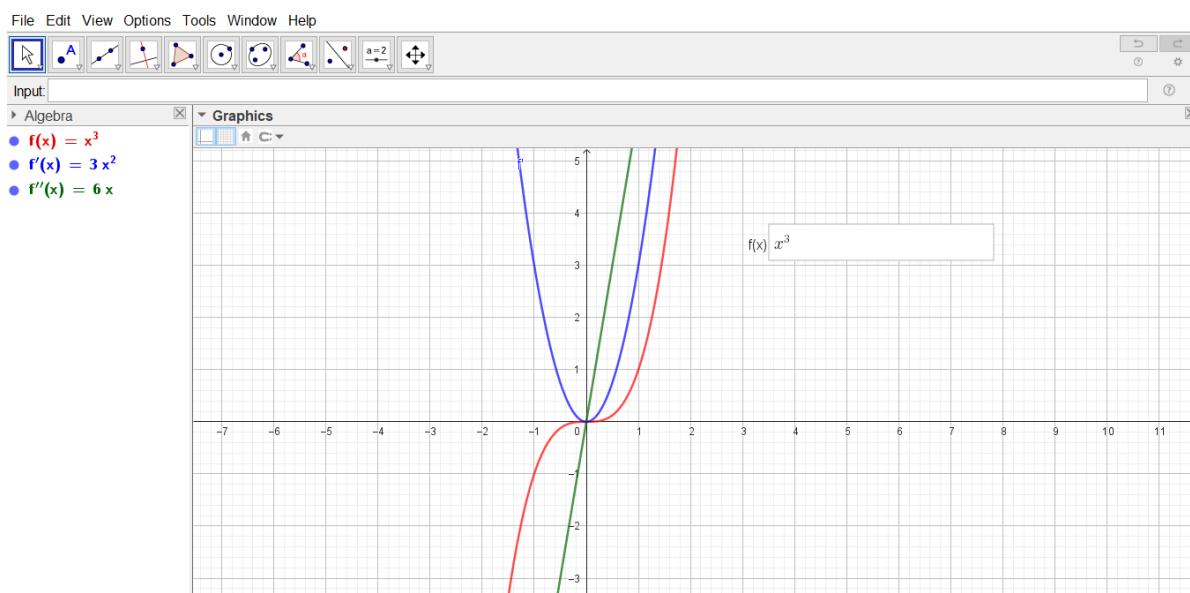
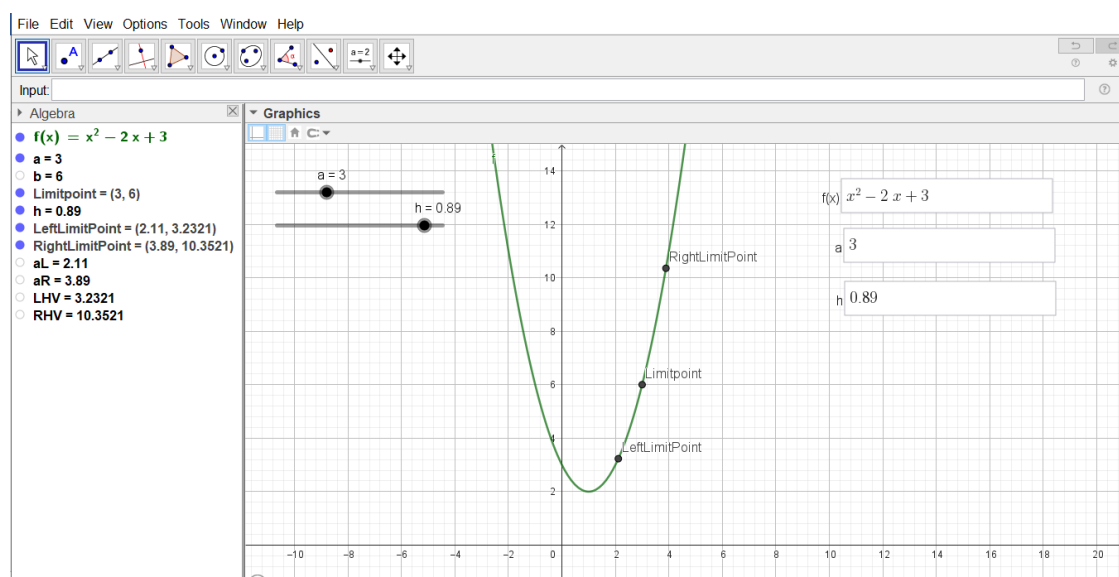
Table-3 Limit of the **Trigonometric** function

h	a-h	a+h	LHV for $\frac{\sin 3x}{2x}$	RHV for $\frac{\sin 3x}{2x}$	LHV for $\frac{\tan 3x}{2x}$	RHV for $\frac{\tan 3x}{2x}$
1	-1	1	0.0706	0.0706	-0.0713	-0.0713
0.8	-0.8	0.8	0.4222	0.4222	-0.5725	-0.5725
0.6	-0.6	0.6	0.8115	0.8115	-3.5719	-3.5719
0.5	-0.5	0.5	0.9975	0.9975	14.1014	14.1014
0.4	-0.4	0.4	1.165	1.165	3.2152	3.2152
0.2	-0.2	0.2	1.4116	1.4116	1.7103	1.7103
0.1	-0.1	0.1	1.4776	1.4776	1.5467	1.5467
0.01	-0.01	0.01	1.4998	1.4998	1.5005	1.5005
			$\lim_{x \rightarrow 0} \frac{\sin 3x}{2x} = \mathbf{1.5}$		$\lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \mathbf{1.5}$	

Table-4 First derivative and Second derivative

Function $f(x)$	First derivative $f'(x)$	Second derivative $f''(x)$
$\sin x$	$\cos x$	$-\sin x$
$\cos x$	$-\sin x$	$-\cos x$
$\tan x$	$\tan^2 x + 1$	$2 \tan x + 2 \tan^3 x$
$\sec x$	$\frac{\sin x}{\cos^2 x}$	$\frac{\cos^2 x + 2 \sin^2 x}{\cos^3 x}$
x^3	$3x^2$	$6x$
-5	0	0
x	1	0
e^x	e^x	e^x

Output:



Exercise No-8

Question

The alternating current passing through a circuit is $i(t) = I_m \sin \omega t$ where, $I_m = 90$ is the maximum value of current and $\omega = 30$ is the angular velocity. Let $L = 0.2$ be the inductance.

- Graph the sinusoidal wave form of $i(t)$ for the given values of I_m and ω .
- Graph the voltage using the formula $v(t) = L \frac{di(t)}{dt}$ for the given value of L .
- Determine the values of $i(t)$ and $v(t)$ for a fixed t and different values of ω .
- Determine the values of $i(t)$ and $v(t)$ for fixed value of ω and different values of t .
- Determine the values of t for which $i(t)$ and $v(t)$ are equal.

Aim:

- To Graph the sinusoidal wave form of $i(t)$ for the given values of I_m and ω .
- To Graph the voltage using the formula $v(t) = L \frac{di(t)}{dt}$ for the given value of L .
- To Determine the values of $i(t)$ and $v(t)$ for a fixed t and different values of ω .
- To Determine the values of $i(t)$ and $v(t)$ for fixed value of ω and different values of t .
- To Determine the values of t for which $i(t)$ and $v(t)$ are equal.

Procedure:

S.No	Description	Geogebra Keywords/Tools
1	Sinusoidal waveform of $i(t)$	Graphics setting $\rightarrow$ Basics $\rightarrow$ Xmin = -1 & Xmax = 1 Ymin = -600 & Ymax = 600
		$i(t) = I_m \sin(\omega t)$ Slider $\rightarrow I_m = 90$, $\omega = 30$ L (Slider) $\rightarrow 0.2$
2	Voltage $v(t)$	$v(t) = L \text{ Derivative}(i)$
3	time	t (slider) Properties $\rightarrow$ min 0, max 10
4	$i(t)$	$I = i(t)$
5	$v(t)$	$V = v(t)$
6	Input box for t and ω	Slider Tool $\rightarrow$ Input box for t and ω
7	Values of t for which $i(t)$ and $v(t)$ are equal	Intersect(i , v , 0, 3)

Output Table-1: Output for a fixed $t = 1$ and different values of ω

ω	$i(t)$	$v(t)$
10	-48.96	-151.03
30	-88.92	83.3
45	76.58	425.51
60	-27.43	-1028.61
90	80.46	-725.88

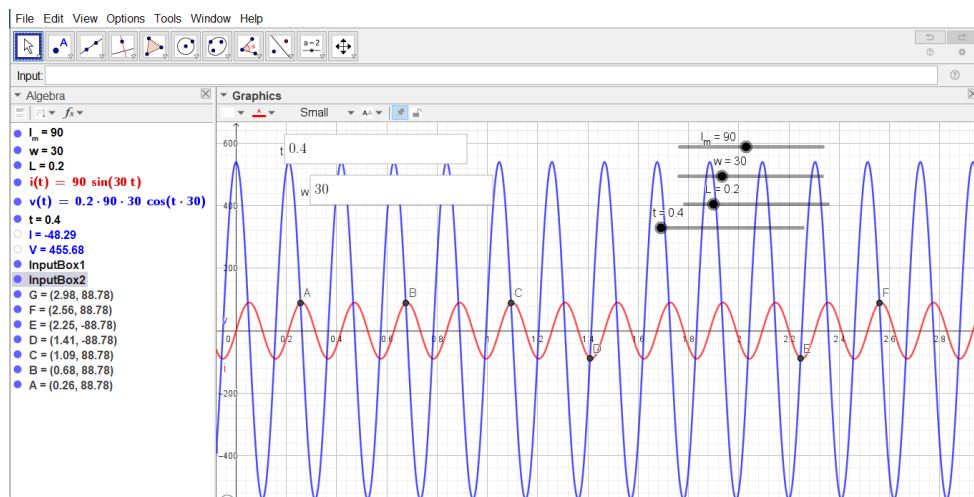
Table-2: Output for a fixed $\omega = 30$ and different values of t

t	$i(t)$	$v(t)$
0	0	0
0.2	-25.15	3.46
0.5	58.53	-6.84
1	-88.92	2.78
2	-27.43	-34.29

Table-3: Output for $i(t) = v(t)$ values and corresponding t values

t	$i(t) = v(t)$
0.26	88.78
0.68	88.78
1.09	88.78
1.41	-88.78
2.25	-88.78
2.56	88.78
2.98	88.78

Output :



Exercise No-9

Question

Do the following activities.

- i. Graph the functions x^3 and $\sin x$. Find their indefinite integrals.
- ii. Evaluate the definite integral $\int_0^5 x \, dx$ and relate it to the area under the curve $y = x$ between x -axis, $x = 0$ and $x = 5$.
- iii. Find the volume of the solid generated by the revolution of the area bounded by $y = x$, x -axis, $x = 0$ and $x = 5$ about x -axis.

Aim

- i. To Graph the functions x^3 and $\sin x$. To Find their indefinite integrals.
- ii. To Evaluate the definite integral $\int_0^5 x \, dx$ and relate it to the area under the curve $y = x$ between x -axis, $x = 0$ and $x = 5$.
- iii. To Find the volume of the solid generated by the revolution of the area bounded by $y = x$, x -axis, $x = 0$ and $x = 5$ about x -axis.

Procedure:

Window-1

S.No	Description	Geogebra Keywords/Tools
1	Function	$f(x) = x^3$
2	Indefinite Integral	Integral(f)
3	Input box	Slider $\rightarrow$ Input box for $f(x)$ Type $f(x) = \sin x$

Window-2

S.No	Description	Geogebra Keywords/Tools
1	Function	$f(x) = x$
2	Definite Integral	Integral(f , a , b) Slider $\rightarrow a = 0$, $b = 5$
3	Area under the curve	Integralbetween(f , $xAxis$, a , b)
4	Input box	Slider $\rightarrow$ Input box for $f(x)$, a , b Type $f(x) = \sin x$

Window-3

S.No	Description	Geogebra Keywords/Tools
1	Function	$f(x) = \text{If}(a \leq x \leq b, x)$
2	Angle α	Slider $\rightarrow$ angle $\rightarrow \alpha$
3	Rotation	$\text{Rotate}(f, \alpha, \text{xAxis})$
4	Surface area	$\text{Surface}(f, \alpha, \text{xAxis})$
5	Volume	$\text{Volume} = \pi \int_a^b f^2(x) dx$
6	Input box	Slider Tool $\rightarrow$ input box $\rightarrow$ for $f(x)$, a , b

Output:**Table-1 :** Output for indefinite integrals

Function $f(x)$	Integral $\int f(x) dx$
5	$5x$
x^3	$\frac{1}{4}x^4$
e^x	e^x
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\sec^2 x$	$\tan x$
$\csc^2 x$	$-\frac{1}{\tan(x)}$
$\sec x \tan x$	$\frac{1}{\cos(x)}$
$\csc x \cot x$	$-\frac{1}{\sin(x)}$

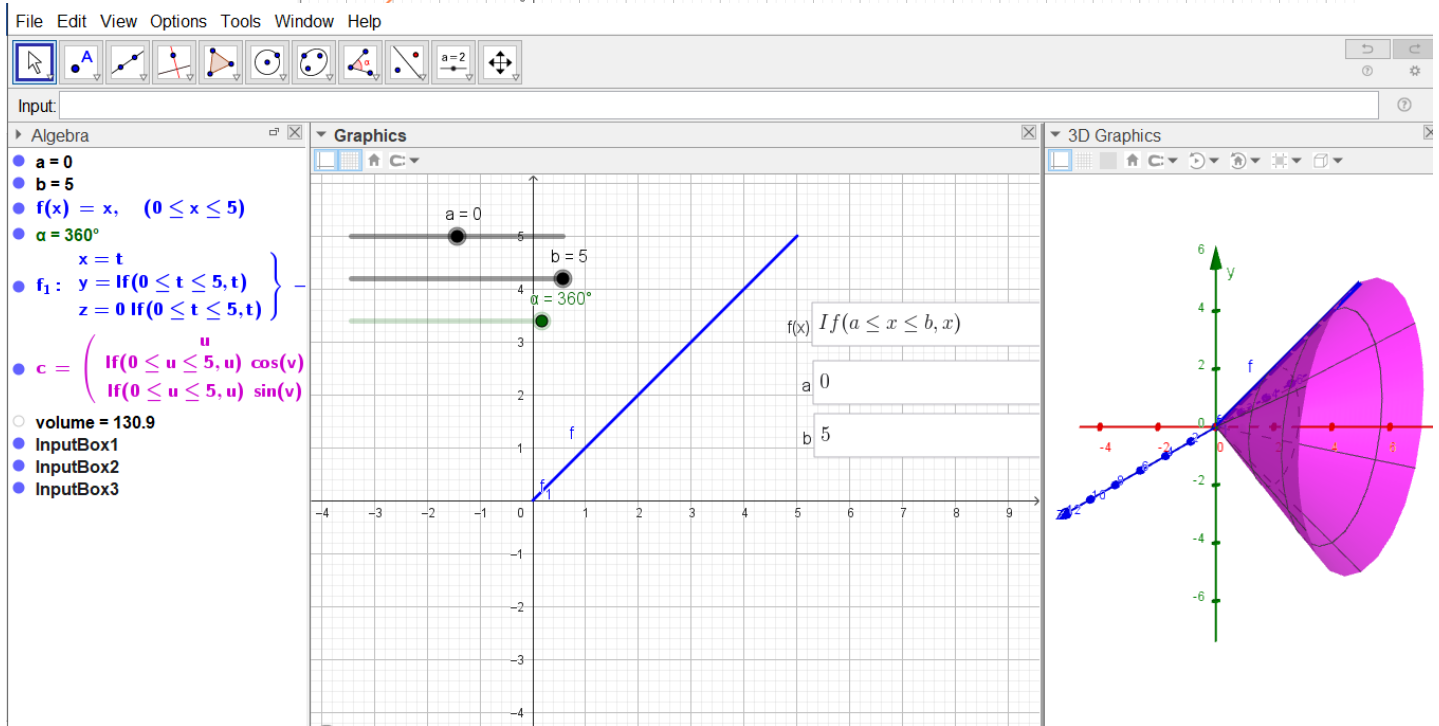
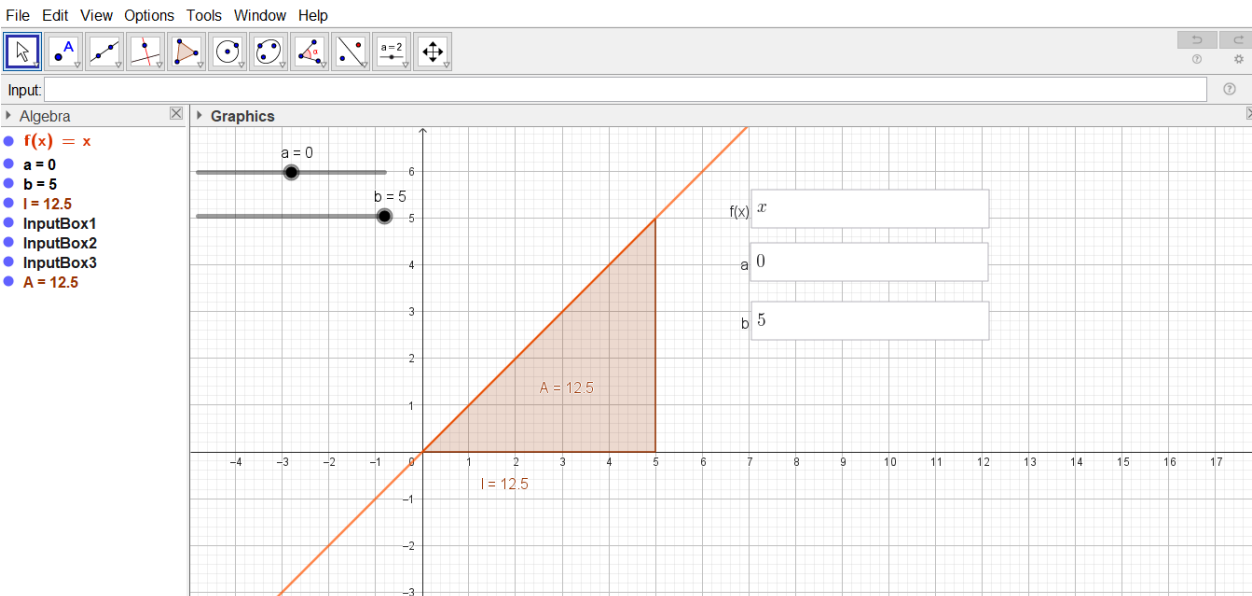
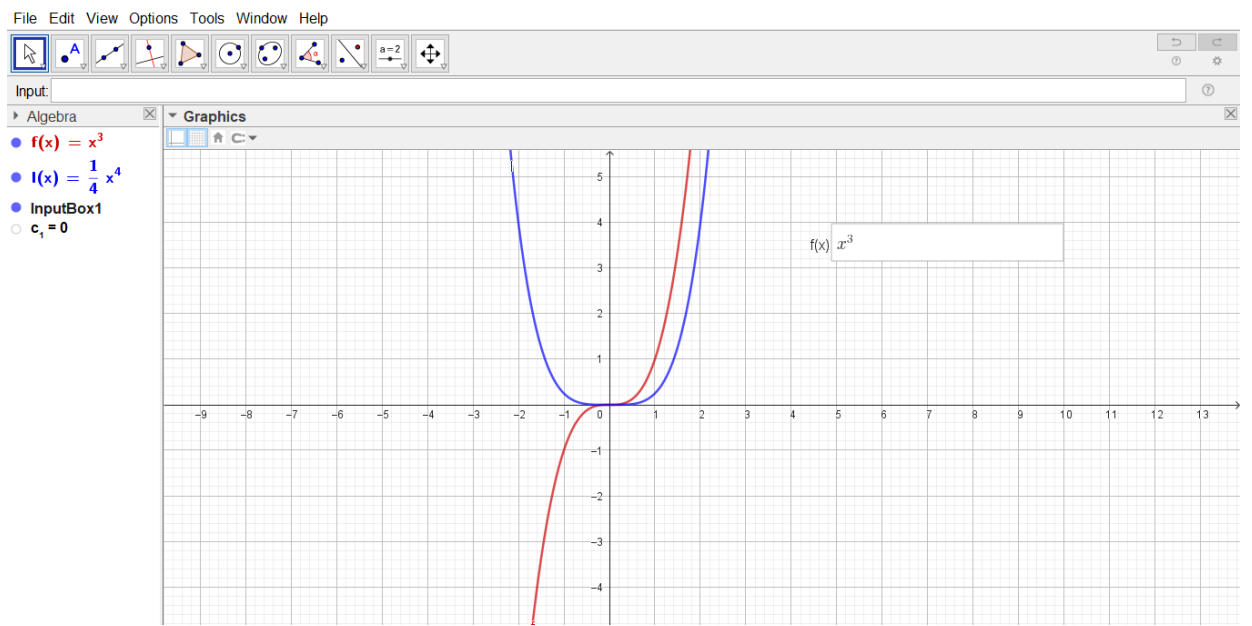
Table-2 : Output for definite integral and area under the curve

Function $f(x)$	$x = a$	$x = b$	$\int_a^b f(x) dx$	Area Between the $f(x)$, x -axis , $x = a$ and $x = b$
x	0	5	12.5	12.5
$\sin x$	0	5	0.72	0.72

Table-3 : Output for the volume of the solid revolution of surface generated by a curve

Function $f(x)$	$x = a$	$x = b$	Volume $= \pi \int_a^b f(x)^2 dx$
x	0	5	130.9
$\sin x$	0	5	8.28

Output :



Exercise No-10

Question

Let $V=20$ be the voltage, $L=2$ be the inductance and $R=5$ be the resistance of a circuit.

- Graph the function of current at time t using the formula $I(t) = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}}\right)$.
- Graph the function of charge passing through the coil at time t using the formula $Q(t) = \int I(t) dt$.
- Find the values of $I(t)$ and $Q(t)$ for given values of V , L and R at different t values.
- Find the values of $I(t)$ and $Q(t)$ at a fixed time for different values of V , L and R .

Aim :

- To Graph the function of current at time t using the formula $I(t) = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}}\right)$.
- To Graph the function of charge passing through the coil at time t using the formula $Q(t) = \int I(t) dt$.
- To Find the values of $I(t)$ and $Q(t)$ for given values of V , L and R at different t values.
- To Find the values of $I(t)$ and $Q(t)$ at a fixed time for different values of V , L and R .

Procedure :

S.No	Description	Geogebra Keywords/Tools
1	$I(t)$	$I(t) = (V / R) (1 - e^{-(R t / L)})$ Slider $\rightarrow V = 20$, $R = 5$, $L = 2$
2	$Q(t)$	$Q(t) = \text{integral}(I(t))$
3	Time t	Slider $\rightarrow t$ Min = 0, Max = 10
4	Current $I(t)$	Current = $I(t)$
5	Charge	Charge = $Q(t)$

Output :

Table-1: Output for I(t) and Q(t) for V = 20 , R = 5 , L = 2

t	I(t)	Q(t)
0	0	1.6
0.5	2.85	2.46
1	3.67	4.13
2	3.97	8.01
4	4	16
8	4	32
10	4	40

Table-2: Output for I(t) and Q(t) for t = 2 and various values of V , R , L

V	R	L	I(t)	Q(t)
10	1	2	6.32	27.36
30	2	4	9.48	41.04
50	3	6	10.54	45.6
70	4	8	11.06	47.88
90	5	10	11.38	49.24

Output :

